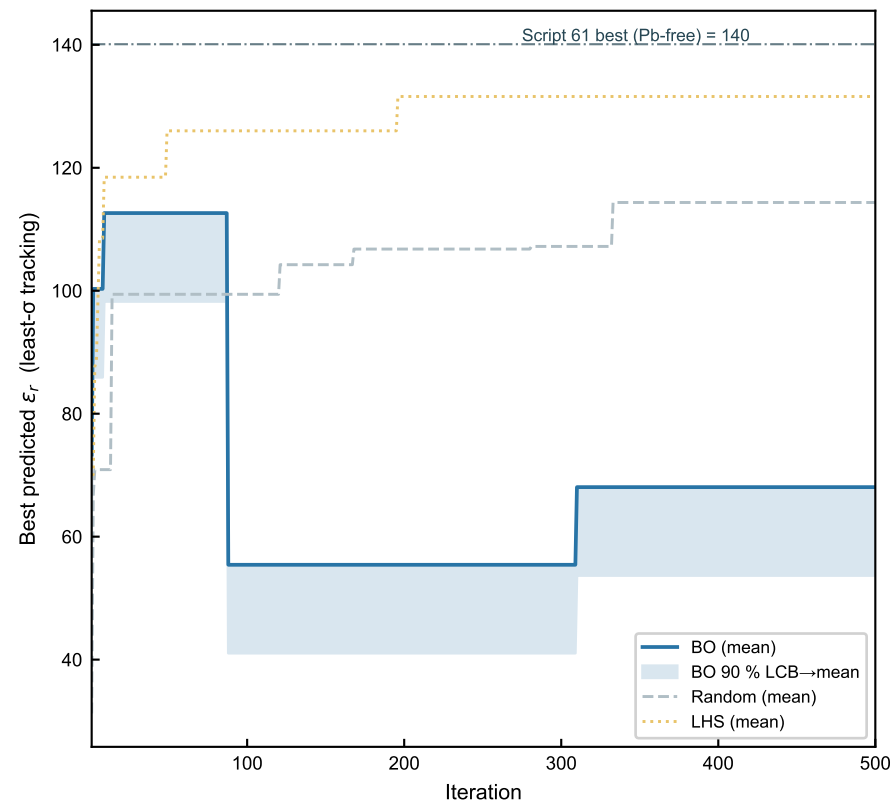
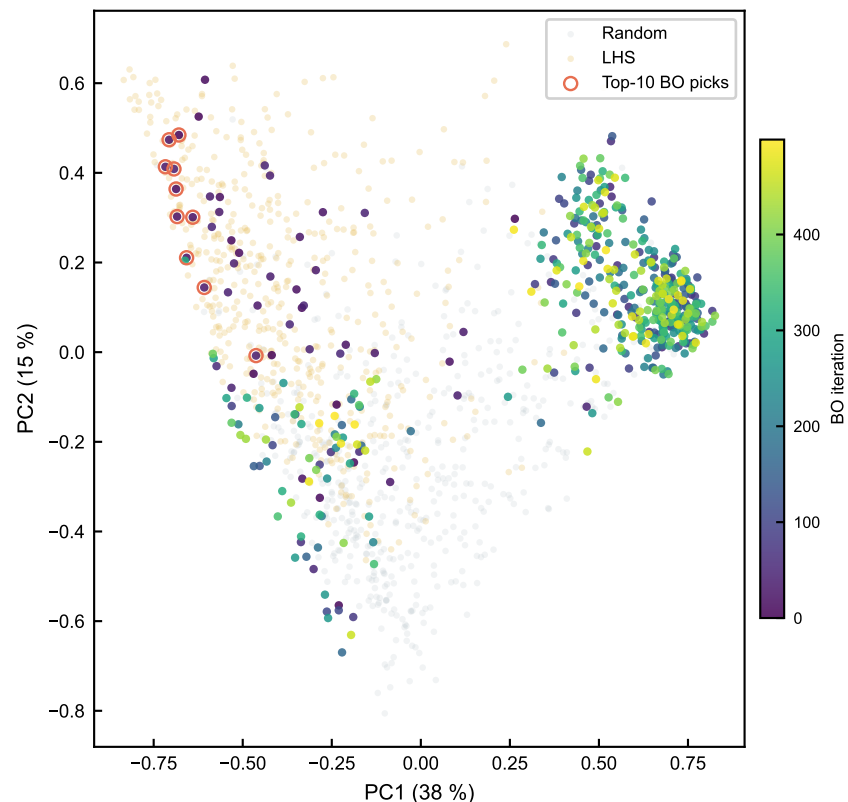
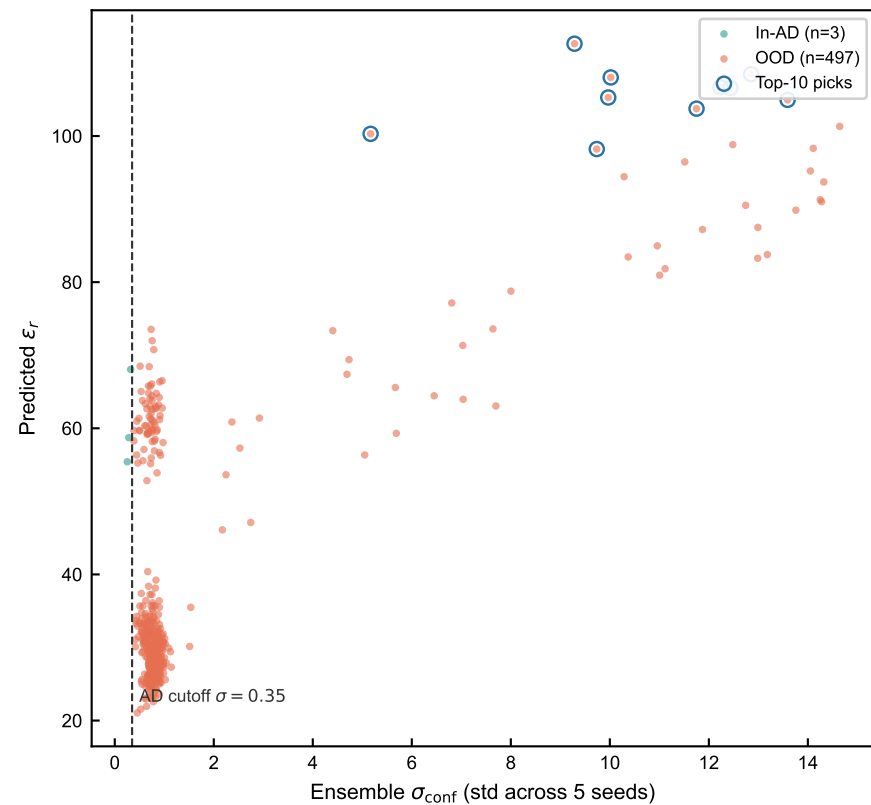


**a Convergence (Pb-free)****b Composition-space PCA (Pb-free)****c BO-selected:  $\varepsilon_r$  vs  $\sigma$** **d Top-10 BO candidates (Pb-free, ranked by 90 % LCB)**

| Composition                    | $\varepsilon_r$ | 90 % CI   | $\sigma$ | A-dom |
|--------------------------------|-----------------|-----------|----------|-------|
| Ba0.57Sr0.3913Ca0.0205Nd0.01.. | 112.6           | [98, 127] | 9.29     | Ba    |
| Ba0.91Sr0.0873Ca0.0016Ti0.75.. | 100.3           | [86, 115] | 5.17     | Ba    |
| Ba0.71Sr0.2117Ca0.065La0.009.. | 108.0           | [94, 122] | 10.02    | Ba    |
| Ba0.81Sr0.1539La0.0075Nd0.00.. | 105.3           | [91, 120] | 9.97     | Ba    |
| Ba0.39Sr0.3233Ca0.2322La0.03.. | 108.5           | [94, 123] | 12.85    | Ba    |
| Ba0.53Sr0.3619Ca0.0595La0.02.. | 106.7           | [92, 121] | 12.23    | Ba    |
| Ba0.55Sr0.1035Ca0.1421La0.13.. | 106.6           | [92, 121] | 12.44    | Ba    |
| Ba0.77Sr0.0989Ca0.0511La0.02.. | 103.7           | [89, 118] | 11.75    | Ba    |
| Ba0.31Sr0.1173Ca0.2119La0.09.. | 98.2            | [84, 113] | 9.74     | Ba    |
| Ba0.35Sr0.3705Ca0.137La0.095.. | 104.9           | [91, 119] | 13.59    | Sr    |